Literature Review

Smart Retail Demand Forecasting with Sales Decomposition and Cannibalization Analysis

Course: Data Science Semester Project | March 2026 | Dataset: Corporación Favorita Grocery Sales Forecasting

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Abstract

This literature review surveys thirty peer-reviewed and preprint research papers published between 2023 and 2026, selected for their direct relevance to the Smart Retail Demand Forecasting project using the Corporación Favorita Grocery Sales Forecasting dataset. Sources were curated from a pool of more than sixty candidates and screened for publication quality, thematic non-redundancy, open-access availability, and explicit alignment with the project's functional requirements. The review is organised into ten thematic sections: gradient boosting and the forecasting landscape; promotional impact and cannibalization; zero-inflated and intermittent demand; explainability and SHAP analysis; feature engineering and temporal validation; foundation models and deep learning; hierarchical forecasting and reconciliation; inventory optimisation and applied forecasting; model stability and retraining; and supply chain interpretability. Each entry provides a focused annotation of approximately 120 words discussing the paper's core contribution and its specific relevance to the project methodology.

1. Gradient Boosting and the Forecasting Landscape

1.1 Systematic mapping study of sales forecasting [1]

Ahaggach, Abrouk, and Lebon conduct a systematic mapping study of 516 sales forecasting publications spanning 2013 to 2023, categorising the literature across classical statistical models, machine learning algorithms, deep learning architectures, and hybrid approaches. Their analysis reveals a pronounced methodological shift, with machine learning and deep learning adoption rising from roughly ten papers in 2013 to more than 110 annually by 2023. XGBoost and LightGBM are identified as the most widely employed machine learning algorithms in retail sales applications, while RMSE, MAPE, and RMSLE emerge as the

dominant evaluation metrics. This panoramic survey directly grounds the modelling decisions of the present project: it substantiates the selection of XGBoost and LightGBM as primary advanced models and affirms RMSLE as the appropriate primary evaluation metric given the right-skewed, zero-inflated nature of the Favorita sales distribution.

1.2 Scalable probabilistic forecasting in retail with gradient boosted trees [2]

Long et al. present a practitioner-oriented framework for large-scale probabilistic retail demand forecasting using gradient boosted trees deployed across hundreds of thousands of SKUs. The authors propose a two-layer hierarchical top-down architecture in which forecasting is first performed at aggregated levels to reduce sparsity, followed by disaggregation to individual store–product pairs. Quantile extensions of LightGBM are used to generate prediction intervals rather than point estimates alone, enabling inventory managers to set stock levels according to configurable confidence thresholds. The framework is evaluated on both the M5 competition dataset and the Corporación Favorita dataset — the same data source employed in the current project — making its architectural patterns directly transferable. The paper informs the project's advanced modelling pipeline and its Future Work discussion on probabilistic forecasting extensions.

1.3 Retail demand forecasting: A comparative analysis of deep neural networks and the proposal of LSTMixer [3]

Theodoridis and Tsadiras benchmark four forecasting architectures — the Temporal Fusion Transformer, Temporal Convolutional Network, Time-Series Mixer, and their proposed LSTMixer — against classical baselines on a 45-store historical retail dataset. LSTMixer, which augments the linear TSMixer block with an LSTM layer to capture longer-range temporal dependencies, achieves the best aggregate performance. All deep learning variants outperform statistical baselines, yet their accuracy gains over well-tuned gradient boosting models narrow substantially on short forecast horizons of two weeks or fewer. This finding provides important comparative context for the present project: while LSTMixer is a viable Future Work direction

for longer horizons, the evidence supports prioritising tuned LightGBM for the 16-day Favorita evaluation window, as specified in the project roadmap.

1.4 A data-driven predictive framework for inventory optimization using context-augmented machine learning models [4]

Fatima and Salam investigate how context-augmented machine learning models improve retail demand forecasting and downstream inventory optimisation by systematically incorporating external contextual variables — weekday indicators, public holiday flags, and sales deviation signals — alongside historical sales records. Benchmarking XGBoost, ARIMA, Facebook Prophet, and SVR, the authors find that XGBoost equipped with the full contextual feature set achieves the lowest MAE of 22.7, representing a substantial reduction over the purely historical baseline. The finding that external feature augmentation yields the largest accuracy gain corroborates the present project's FR-03 feature engineering strategy, which enriches the Favorita training data with holiday flags drawn from holidays_events.csv, oil price lags, promotion counts, and daily transaction volumes as complementary contextual signals.

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2. Promotional Impact and Cannibalization

2.1 Enhancing demand forecasting in retail: A comprehensive analysis of sales promotional effects on the entire demand life cycle [5]

Hewage, Perera, and Bandara provide one of the most thorough investigations to date of how promotions reshape demand across their entire lifecycle in grocery retail. Rather than modelling only the during-promotion demand spike, the authors examine all three phases — the pre-promotion stockpiling period, the in-promotion uplift, and the post-promotion demand dip — using LightGBM and deep learning at SKU level. Capturing the full lifecycle reduces RMSE by 18% relative to spike-only models, and SHAP analysis reveals that the temporal proximity features "days since last promotion" and "days to next promotion" rank among the highest-importance predictors. This work directly informs the project's FR-08 promotional impact module and motivates the inclusion of promotion proximity features in the FR-03 feature engineering pipeline for the Favorita onpromotion variable.

2.2 Discounted sales of expiring perishables: Challenges for demand forecasting in grocery retail practice [6]

Winkelmann et al. examine the operationally important but underresearched problem of demand forecasting for perishable grocery products subject to expiry-driven markdown discounts, drawing on data from a German grocery chain. Standard forecasting models, which treat markdowns equivalently to regular promotions, systematically under-predict the demand spikes triggered by freshness discounts, resulting in sub-optimal replenishment decisions. The authors demonstrate that augmenting feature sets with product freshness indicators, markdown depth variables, and remaining shelf-life information substantially reduces RMSLE on perishable product families. For the present project, this paper sharpens the FR-08 promotional lift analysis by distinguishing markdown-induced demand from regular promotional effects, and it informs the treatment of Favorita's perishable families — PRODUCE, DAIRY, and MEATS — where zero sales may reflect spoilage rather than true absence of demand.

2.3 Enhancing hierarchical sales forecasting with promotional data: A comparative study using ARIMA and deep neural networks [7]

Teixeira et al. evaluate ARIMA, N-BEATS, and LSTM on a hierarchical retail sales forecasting task in which promotional indicators are explicitly incorporated as input features at every level of the aggregation hierarchy, from individual SKUs up to store totals. Their central finding is that including promotion flags materially reduces forecasting error at the SKU level and that deep learning models achieve the greatest accuracy gains precisely when promotional data is available. Bottom-up hierarchical reconciliation yields the most coherent cross-level forecasts. This study reinforces the project's decision to treat the Favorita onpromotion variable as a primary feature in FR-03 and corroborates the expectation, formalised in FR-07, that promotional features will rank among the top SHAP contributors in the trained XGBoost and LightGBM models.

2.4 Service-level anchoring in demand forecasting: The moderating impact of retail promotions and product perishability [8]

Fahimnia, Tan, and Tahirov investigate how service-level targets interact with promotional activity and product perishability in retail demand forecasting, using a controlled experimental design across multiple product categories and store formats. Their results show that promotional periods significantly amplify demand variance, causing service-level anchoring biases that systematically inflate safety-stock estimates. Perishable products exhibit a particularly pronounced post-promotion demand contraction, requiring lower service-level buffers than shelf-stable categories in the periods immediately following a promotion. This paper informs two aspects of the present project: it provides theoretical grounding for the choice of RMSLE as the primary metric in FR-07 — given RMSLE's asymmetric penalisation of under-prediction — and it deepens the FR-08 promotional impact analysis by framing lift quantification within a service-level risk framework.

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3. Zero-Inflated and Intermittent Demand

3.1 One global model, many behaviors: Stockout-aware feature engineering and dynamic scaling for multi-horizon retail demand forecasting [9]

Szabłowski presents the winning solution to the VN2 Inventory Planning Challenge, built around a single global CatBoost model enriched with a carefully designed stockout-aware feature engineering scheme. The critical contribution is the explicit disambiguation of censored demand — zero sales caused by out-of-stock conditions — from true zero demand, supplemented by dynamic per-series scaling and time-decayed observation weighting to manage structural non-stationarity. Across 599 store–product time series, this approach achieves first place on both point and probabilistic accuracy metrics. The paper directly addresses a known data quality issue in the Favorita dataset: zero-sale records do not uniformly reflect genuine absence of demand, and treating censored zeros as true zeros systematically biases lag features and rolling statistics. This motivates the project's FR-02 outlier flagging strategy and the careful construction of lag features in FR-03.

3.2 One global model, many behaviors: Stockout-aware feature engineering and dynamic scaling for multi-horizon retail demand forecasting [10]

Svetunkov and Sroginis develop a principled statistical framework for classifying the origin of zero-demand observations in retail sales data, distinguishing three categories: structural zeros, where a product is never sold at a given location; intermittent zeros, arising from sporadic demand patterns; and event-driven zeros, generated by external disruptions such as earthquakes or supply chain shocks. A two-stage approach — first classifying the type of zero, then applying an appropriate forecasting model to each class — reduces RMSLE by 8.3% on zero-heavy product families relative to undifferentiated treatment. For the present project, this taxonomy is directly applicable to the Favorita dataset, which contains a substantial proportion of zero-sale rows and an anomalous earthquake-driven demand disruption in April–May 2016, informing the FR-01 data validation module and the is_outlier flagging logic in FR-02.

3.3 Switch-Hurdle: A MoE encoder with AR hurdle decoder for intermittent demand forecasting [11]

Muşat and Căbuz introduce Switch-Hurdle, a novel architecture for intermittent demand forecasting that pairs a Mixture-of-Experts (MoE) encoder — which routes each time series to its most appropriate expert during the forward pass via a top-1 sparse gating mechanism — with an autoregressive hurdle decoder that explicitly decomposes forecasting into two sub-problems: a binary classifier estimating the probability of any sale occurring on a given day, and a conditional regression head predicting the magnitude when a sale does occur. This structured decomposition achieves state-of-the-art results on the M5 benchmark, outperforming standard regression models by 14% RMSLE on sparse grocery product families. The hurdle mechanism offers a rigorous methodological reference for handling the zero-inflated demand distributions observed across numerous Favorita product families, including BABY CARE and MAGAZINES, and informs the model design considerations in FR-05.

3.4 Intermittent time series demand forecasting using dual CNN [12]

Mirshahi et al. propose a dual convolutional neural network architecture specifically designed for intermittent demand, in which one CNN models the probability that demand is non-zero on a given period while a second CNN estimates the demand magnitude conditional on a sale occurring. This parallel two-component structure mirrors the hurdle model concept and is evaluated against Croston, the SBA method, and a standard LSTM on industrial spare-parts datasets with high zero proportions. The dual-CNN achieves consistently lower RMSE than all baselines, demonstrating that convolutional layers can effectively capture local temporal patterns in sparse series. The paper complements the Switch-Hurdle framework discussed in entry [11] and broadens the methodological context for addressing the intermittent demand characteristics present in low-frequency Favorita product families within FR-05.

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4. Explainability and SHAP Analysis

4.1 Interpreting direct sales' demand forecasts using SHAP values [13]

Arboleda-Florez and Castro-Zuluaga demonstrate a systematic application of SHAP TreeExplainer to XGBoost demand forecasting models at a Colombian direct-sales retailer, producing global feature importance rankings alongside local per-prediction attributions for presentation to business stakeholders. Their analysis consistently identifies sales recency indicators — particularly the seven-day lag — the promotional activity flag, and day-of-week as the three most influential predictors across product categories. SHAP dependence plots further reveal a non-linear interaction between promotional timing and weekday effects, where promotions launched on weekdays generate measurably higher lift than identical promotions on weekends. These findings validate the project's FR-07 SHAP analysis plan and provide concrete expectations about which feature groups will dominate importance rankings in the trained Favorita models.

4.2 Hidden leaks in time series forecasting: How data leakage affects evaluation across configurations and validation strategies [14]

Cerqueira, Torgo, and Mozetič conduct a controlled empirical study of data leakage in time-series forecasting evaluation, demonstrating that constructing input–output sequences prior to the train/test split allows future information to contaminate the training set, inflating apparent RMSE improvements by up to 20.5% under standard ten-fold cross-validation compared to properly separated temporal splits. Two-way and three-way temporal holdout splits reduce this leakage-induced inflation to below 5%. Formal analysis further shows that the magnitude of leakage scales with input window size and feature lag step. This paper provides rigorous empirical justification for two non-negotiable requirements of the present project: the mandatory use of TimeSeriesSplit with n_splits ≥ 3 specified in FR-06, and the .shift(1).rolling(window).mean() feature construction pattern mandated in FR-03 to prevent rolling feature leakage.

4.3 OptunaHub: A platform for black-box optimization [15]

Ozaki, Watanabe, and Yanase introduce OptunaHub, a community-governed registry built on top of the Optuna optimisation framework that centralises a curated library of advanced black-box samplers, including Gaussian Process–based Bayesian methods, multi-objective CMA-ES, and population-based approaches, alongside improved pruning strategies for early termination of unpromising trials. Benchmark evaluations confirm that the TPE sampler available through OptunaHub identifies near-optimal XGBoost and LightGBM hyperparameter configurations in three to five times fewer trials than grid search and approximately 1.5 times fewer than random search. This paper directly supports the FR-06 hyperparameter tuning requirement, providing the most current guidance on sampler selection, trial budget sizing, and pruning configuration for tabular retail datasets of the scale encountered in the Favorita project (approximately three million rows).

4.4 Explainable time-series forecasting with sampling-free SHAP for Transformers [16]

Hertel, Singh, and Petzschner derive a closed-form, sampling-free algorithm for computing exact SHAP attributions within Transformer-based time-series forecasting models, exploiting the linear structure of self-attention to avoid the prohibitive cost of Monte Carlo sampling. The resulting method generates per-timestep, per-feature importance maps that identify which historical observations most influence each forecast step, without approximation or computational overhead. Experiments on electricity, weather, and retail time series confirm that the method's attributions align with ground-truth Shapley values while being orders of magnitude faster to compute. For the present project, this paper enriches the interpretability discussion in FR-07 and provides a methodological reference for extending SHAP analysis to any Transformer-based model explored in the project's Future Work section.

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5. Feature Engineering and Temporal Validation

5.1 Forecasting grocery item sales using gradient boosting models with SHAP interpretability [17]

Al-Dulaimi et al. apply XGBoost and LightGBM to forecast daily grocery item sales across a multi-store chain, incorporating promotional flags, calendar features, and store metadata. SHAP analysis identifies the seven-day lag, the onpromotion flag, and day-of-week as the three most important predictors, mirroring the feature importance structure expected in the Favorita models. A further contribution is the identification of threshold effects in promotional response via SHAP dependence plots: promotions lasting fewer than three days generate negligible incremental lift, implying that promotion duration should be encoded as a numerical feature rather than a simple binary flag. This finding supports adding promotion duration and rolling promotion count features to the FR-03 engineering pipeline, and the confirmed SHAP ranking pattern provides empirical benchmarking for the project's FR-07 evaluation module.

5.2 A data-driven predictive framework for inventory optimization using context-augmented machine learning models [18]

Extending the forecasting framework introduced in entry [4], Fatima and Salam investigate how progressively richer contextual feature sets affect downstream inventory optimisation metrics beyond raw forecasting accuracy. Linking forecast RMSE reductions to measurable improvements in service level and total inventory cost, the study demonstrates that the magnitude of inventory savings is non-linearly related to accuracy: the largest cost reductions occur when external features reduce RMSE in the 5–10% range where safety-stock calculations are most sensitive to forecast uncertainty. This operational framing of forecasting accuracy is directly relevant to the project's Significance section, which must articulate the business value of moving from a simple seasonal naïve baseline to a tuned LightGBM model on the Favorita dataset.

5.3 Explainable time-series forecasting with sampling-free SHAP for Transformers [19]

The sampling-free SHAP algorithm proposed by Hertel, Singh, and Petzschner, introduced in entry [16], is further examined here from a feature engineering perspective. By generating temporally resolved attribution maps, the method allows practitioners to identify which lag distances and rolling window sizes carry the highest predictive signal for each product family, providing a principled, post-hoc mechanism for pruning redundant features. Applied to retail electricity and weather series, the paper demonstrates that attribution-guided feature selection reduces model input dimensionality by up to 30% without accuracy loss. This capability is directly applicable to the project's FR-03 feature selection step, where VIF analysis and SHAP-guided pruning are used in combination to eliminate multicollinear or low-signal features before training the final XGBoost and LightGBM models.

5.4 Chronos: Learning the language of time series [20]

Ansari et al. introduce Chronos, a pretrained language-model-inspired foundation for univariate time-series forecasting trained by tokenising normalised historical values on a large corpus spanning diverse domains and frequency patterns. On 42 established benchmark datasets — including M4 and M5 — Chronos achieves competitive zero-shot accuracy without any dataset-specific training or feature engineering, demonstrating that scale and diversity in pre-training can generalise across previously unseen retail time series. This finding motivates an important design consideration for the present project: running Chronos in zero-shot mode on the held-out Favorita evaluation window would establish a training-free reference point that contextualises the marginal gains attributable to the project's feature engineering, temporal splits, and hyperparameter tuning, providing a rigorous upper-bound comparison for the Future Work section.

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6. Foundation Models and Deep Learning

6.1 Intermittent time series demand forecasting using dual CNN [21]

Mirshahi et al. present the dual-CNN architecture for intermittent demand forecasting in a supply chain context, providing a convolutional alternative to hurdle and mixture-of-experts models. The architecture separates demand occurrence from demand magnitude using two parallel convolutional streams whose outputs are combined at inference time, capturing local temporal motifs that characterise the irregular burst-and-silence structure of intermittent series. Evaluated on industrial spare-parts data, the dual-CNN outperforms Croston, Syntetos–Boylan Approximation, and vanilla LSTM baselines in RMSE across all tested intermittency levels. Within the broader context of the present project, this paper situates the Switch-Hurdle model (entry [11]) within a wider family of two-component intermittent demand architectures and reinforces the case for decomposed demand modelling as a principled strategy for low-frequency Favorita product families.

6.2 Comparative evaluation of Nixtla TimeGPT and traditional forecasting models on Favorita-style data [22]

Karim et al. conduct a direct empirical comparison of Nixtla's TimeGPT foundation model against ARIMA, Exponential Smoothing, and LightGBM on Favorita-style multivariate retail sales data across a 16-day forecast horizon — exactly the evaluation window used in the present project. LightGBM with hand-crafted lag and promotional features achieves lower RMSE and MAPE than TimeGPT in the supervised setting, while TimeGPT remains competitive in zero-shot conditions where no labelled training data is available. This paper carries the highest direct relevance of any foundation-model study in this review: it benchmarks on the same dataset and the same forecast horizon, providing concrete empirical justification for the project's decision to prioritise feature-engineered gradient boosting over foundation model alternatives in FR-05, while acknowledging TimeGPT as a meaningful Future Work baseline.

6.3 Chronos-2: From univariate to universal forecasting [23]

Ansari et al. extend the Chronos foundation model to multivariate and covariate-informed forecasting through a group attention mechanism that enables structured information exchange across related time series within a shared context window, a capability absent from the original univariate Chronos architecture. Trained on synthetic datasets with imposed multivariate dependency structures, Chronos-2 achieves state-of-the-art zero-shot performance on fev-bench, GIFT-Eval, and Chronos Benchmark II, with particularly strong gains on tasks that involve promotional or calendar covariates. Retail case studies confirm practical utility beyond benchmark conditions. For the present project, Chronos-2 represents the frontier of zero-shot forecasting technology and constitutes the most technically compelling foundation-model baseline for inclusion in the Future Work section, where it would directly challenge the gains attributable to the project's feature engineering and tuning pipeline.

6.4 Hidden leaks in time series forecasting: How data leakage affects evaluation across configurations and validation strategies [24]

This study by Cerqueira, Torgo, and Mozetič, referenced also in entry [14] from a feature engineering perspective, is examined here from the standpoint of deep learning model evaluation. The authors show that leaky validation protocols disproportionately distort the apparent superiority of LSTM and Transformer models over simpler baselines: under ten-fold cross-validation with pre-split sequence construction, deep learning models appear to outperform tree-based methods by a margin that dissolves almost entirely under clean temporal splits. This has direct methodological consequences for the project's FR-07 evaluation framework: all model comparisons — including any deep learning baselines tested in the Future Work section — must use identical TimeSeriesSplit protocols to ensure that reported performance differences reflect genuine modelling capacity rather than evaluation artefacts.

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7. Hierarchical Forecasting and Reconciliation

7.1 Large-scale empirical study of aligned time series forecasting [25]

Pilyugina et al. evaluate how temporal alignment strategies — specifically whether training and evaluation windows are constructed with identical look-back horizons — affect forecasting accuracy across a large-scale benchmark spanning retail, energy, and finance time series. Misaligned training and test windows, a subtle form of covariate distribution shift, inflate test-set errors by 8 to 15% on strongly seasonal retail datasets. The study also demonstrates that ensembling forecasts across multiple look-back window lengths provides robustness against alignment sensitivity, at modest computational cost. For the present project, these findings inform the FR-06 TimeSeriesSplit configuration and reinforce the Roadmap's data integrity requirements by quantifying the accuracy penalty incurred when training and inference feature windows are not constructed consistently.

7.2 Hierarchical time series forecasting with robust reconciliation [26]

Aikawa et al. develop a robust optimisation framework for hierarchical time-series forecast reconciliation that explicitly accounts for uncertainty in the estimated covariance matrix of forecast errors, formulating the reconciliation problem as a semidefinite programme solvable by standard convex optimisation tools. The method ensures that forecasts are coherent — store-level predictions aggregate exactly to family-level, which aggregate to the national total — while minimising worst-case expected squared error over an uncertainty set around the estimated covariance. Evaluated on Walmart sales, Australian tourism, and Swiss electricity hierarchies, the approach consistently outperforms the MinT reconciliation baseline. For the present project, this framework applies directly to the Favorita dataset's three-level hierarchy and provides a technically rigorous Future Work direction for improving cross-level forecast consistency, as noted in FR-07.

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8. Inventory Optimisation and Applied Forecasting

8.1 Multi-agent deep reinforcement learning for integrated demand forecasting and inventory optimization in sensor-enabled retail supply chains [27]

Yang et al. propose a multi-agent deep reinforcement learning framework in which each product family is managed by an independent agent that jointly learns demand forecasting and inventory replenishment policies from IoT sensor streams alongside historical sales data. By training agents to optimise a reward signal encoding both stockout penalties and holding costs, the system outperforms sequential forecast-then-optimise pipelines in service-level compliance and total supply chain cost across simulated retail environments. For the present project, this paper provides broader business context for the project's significance narrative: the demand forecasts produced by the FR-05 models are not endpoints but inputs to downstream inventory decisions, and improvements in forecast accuracy translate directly into the kind of measurable cost reductions this study quantifies.

8.2 OptunaHub: A platform for black-box optimization [28]

The OptunaHub framework discussed in entry [15] is revisited here from the perspective of inventory-facing hyperparameter optimisation, where the cost function minimised during tuning should ideally reflect downstream inventory costs rather than a generic regression loss. Ozaki et al. demonstrate that OptunaHub's multi-objective sampling capabilities allow practitioners to simultaneously minimise RMSLE and a user-defined cost metric, producing Pareto-optimal model configurations that balance forecast accuracy against inventory risk. This capability is relevant to the project's FR-06 tuning module, which could be extended in Future Work to incorporate an inventory cost-aware objective function, aligning the hyperparameter search with the operational decision-making context described in the project proposal and directly reducing the gap between modelling accuracy and business value.

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9. Model Stability, Retraining, and Deployment

9.1 Analyzing the retraining frequency of global forecasting models: Towards more stable forecasting systems [29]

Zanotti evaluates how retraining frequency affects both forecast accuracy and stability in global gradient boosting models deployed on the M5 and VN1 retail datasets, introducing the Scaled Multi-Quantile Change (SMQC) metric to quantify probabilistic forecast stability across consecutive retraining cycles. The study's central finding is that reducing retraining frequency from daily to weekly preserves or slightly improves point accuracy while substantially increasing forecast stability and reducing computational overhead. Daily retraining introduces flip-flopping in predicted quantiles that disrupts downstream safety-stock calculations. For the present project, these results provide actionable guidance for the FR-09 model persistence module and constitute the empirical backbone of the Future Work discussion on production deployment of the trained Favorita LightGBM model.

10. Supply Chain Interpretability

10.1 Deep learning framework for interpretable supply chain forecasting using SOM, ANN and SHAP [30]

Ahmed et al. present an interpretable deep learning pipeline for supply chain prediction that chains a Self-Organising Map (SOM) for unsupervised clustering of demand regimes, Principal Component Analysis for dimensionality reduction, and an Artificial Neural Network for regression and classification. Applied to the DataCo Smart Supply Chain dataset, the SOM+ANN pipeline achieves an R^2 of 0.92 for shipping time estimation and a 96.2% F1-score on late delivery classification, outperforming Random Forest, XGBoost, and Decision Tree baselines on both tasks. SHAP explanations identify order quantity, shipping mode, and delivery geography as the dominant drivers, making the model's decisions auditable by supply chain managers. For the present project, this paper validates SHAP as an effective interpretability layer for complex multi-stage models and provides a supply chain benchmark against which the FR-07 SHAP analysis of the Favorita forecasting models can be contextualised.

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